THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 13: **Statistics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of a distribution with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 13, 11, 2, p, 14, 5 respectively is 37.2. The value of p is: 1

(a) 4 **(b)** 5 **(c)** 7 **(d)** 10

2. The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80 and frequencies 7, x, 16, y, 3, 4 respectively, and total frequency 50, is 45. The value of x is: 1

(a) 9 **(b)** 12 **(c)** 11 **(d)** 10

3. The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 7, 14, x, 20, y, 19, 3 respectively, and total frequency 100, is 54. The value of y is: 1

(a) 18 **(b)** 20 **(c)** 22 **(d)** 16

4. The median class of a distribution is the class whose cumulative frequency is: 1

(a) the highest frequency **(b)** just less than $\frac{N}{2}$

(c) just greater than $\frac{N}{2}$ **(d)** equal to N always

5. For the data with classes 70–80, 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 with frequencies 8, 19, 19, 20, 16, 9, 9 respectively, The mean of the data is: 1

(a) 104 **(b)** 102 **(c)** 105 **(d)** 103

6. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 5, 5, 7, 6, 9, 8 respectively, The median of the data is: 1

(a) 50 **(b)** 43.25 **(c)** 45 **(d)** 40

7. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 with frequencies 5, 7, 4, 20, 11, 3 respectively, The mode of the data is: 1

(a) 71.8 **(b)** 70 **(c)** 76.4 **(d)** 80

8. The mean of a distribution with classes 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 19, p, 5, 15, 22, 19 respectively is 60.8. The value of p is: 1

(a) 19 **(b)** 20 **(c)** 22 **(d)** 25

9. The median of a distribution with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140, 140–150 and frequencies 20, x, 16, y, 4, 11, 14 respectively, and total frequency 100, is 105. The value of x is: 1

(a) 21 **(b)** 22 **(c)** 23 **(d)** 15

10. The median of a distribution with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140, 140–150 and frequencies 3, 8, 16, x, 20, y, 10 respectively, and total frequency 100, is 120.5. The value of y is: 1

(a) 22 **(b)** 20 **(c)** 23 **(d)** 21

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Number of workers	8	2	3	20	3	3	11

Find the median.

12. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	70–80	80–90	90–100	100–110	110–120	120–130
Number of workers	5	13	5	2	4	11

Find the mode.

13. The ages (in years) of patients admitted to a hospital in a month are given below. 2

Age (years)	70–80	80–90	90–100	100–110	110–120	120–130	130–140
Number of patients	8	19	19	20	16	9	9

Find the mean.

14. The ages (in years) of patients admitted to a hospital in a month are given below. 2

Age (years)	40–50	50–60	60–70	70–80	80–90	90–100
Number of patients	4	4	15	6	8	3

Find the median.

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	10–20	20–30	30–40	40–50	50–60	60–70
Number of patients	5	5	7	6	9	8

Find the mean.

16. The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. 3

Lifetime (hours)	70–80	80–90	90–100	100–110	110–120	120–130	130–140
Number of bulbs	8	19	19	20	16	9	9

Find the median.

17. The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	10–20	20–30	30–40	40–50	50–60	60–70
Number of patients	13	11	2	5	14	5

Find the mean and the mode (in this order).

OR

The weights (in kg) of students of a class are given below. 3

Weight (kg)	40–50	50–60	60–70	70–80	80–90	90–100
Number of students	8	14	22	20	17	19

Find the mean.

18. The distances (in km) travelled to office by employees of a company are given below. 3

Distance (km)	80–90	90–100	100–110	110–120	120–130	130–140
Number of employees	10	3	3	7	20	7

Find the median.

OR

The weights (in kg) of students of a class are given below. 3

Weight (kg)	30–40	40–50	50–60	60–70	70–80	80–90
Number of students	19	20	5	15	22	19

Find the mean and the mode (in this order).

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. The mean of a distribution with classes 70–80, 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 and frequencies 8, 19, 19, 20, p, 9, 9 respectively is 103. Find the value of p. 5

OR

The median of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 and frequencies 6, 14, x, 18, y, 18, 14 respectively, and total frequency 100, is 75. Find the value of x. 5

20. The mean of a distribution with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 and frequencies 10, p, 3, 7, 20, 7 respectively is 114. Find the value of p. 5

OR

The median of a distribution with classes 30–40, 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 and frequencies 4, 9, x, 10, y, 6, 6 respectively, and total frequency 50, is 69. Find the value of x. 5

END OF QUESTION PAPER Total: 40 Marks